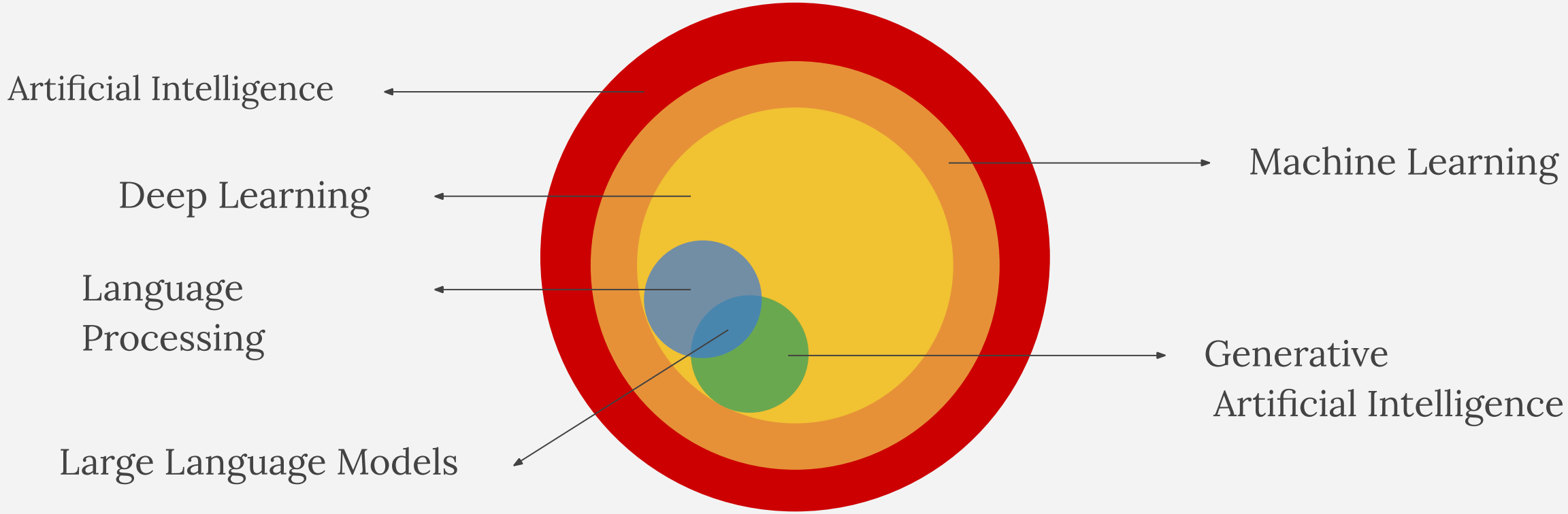
Large Language Models: High-Level View

Yasaman Boreshban

Most slides have been adopted from Andrej Karpathy

AI, ML, DL, Gen AI, NLP, LLM?



What is AI?

AI Effect: *It's part of the history of the field of artificial intelligence that every time somebody figured out how to make a computer do something—play good checkers, solve simple but relatively informal problems—there was a chorus of critics to say, 'that's not thinking*

“***AI is whatever hasn't been done yet***”



What is Machine Learning?

Learning From Experience

Elements of Learning:

- Data
- Architecture and Parameters
- Objective
- Training

What is Deep Learning?

Neural Networks: A special type of architecture in machine learning models

Not New Idea, jump after 2012 due to the advancement of hardware and data

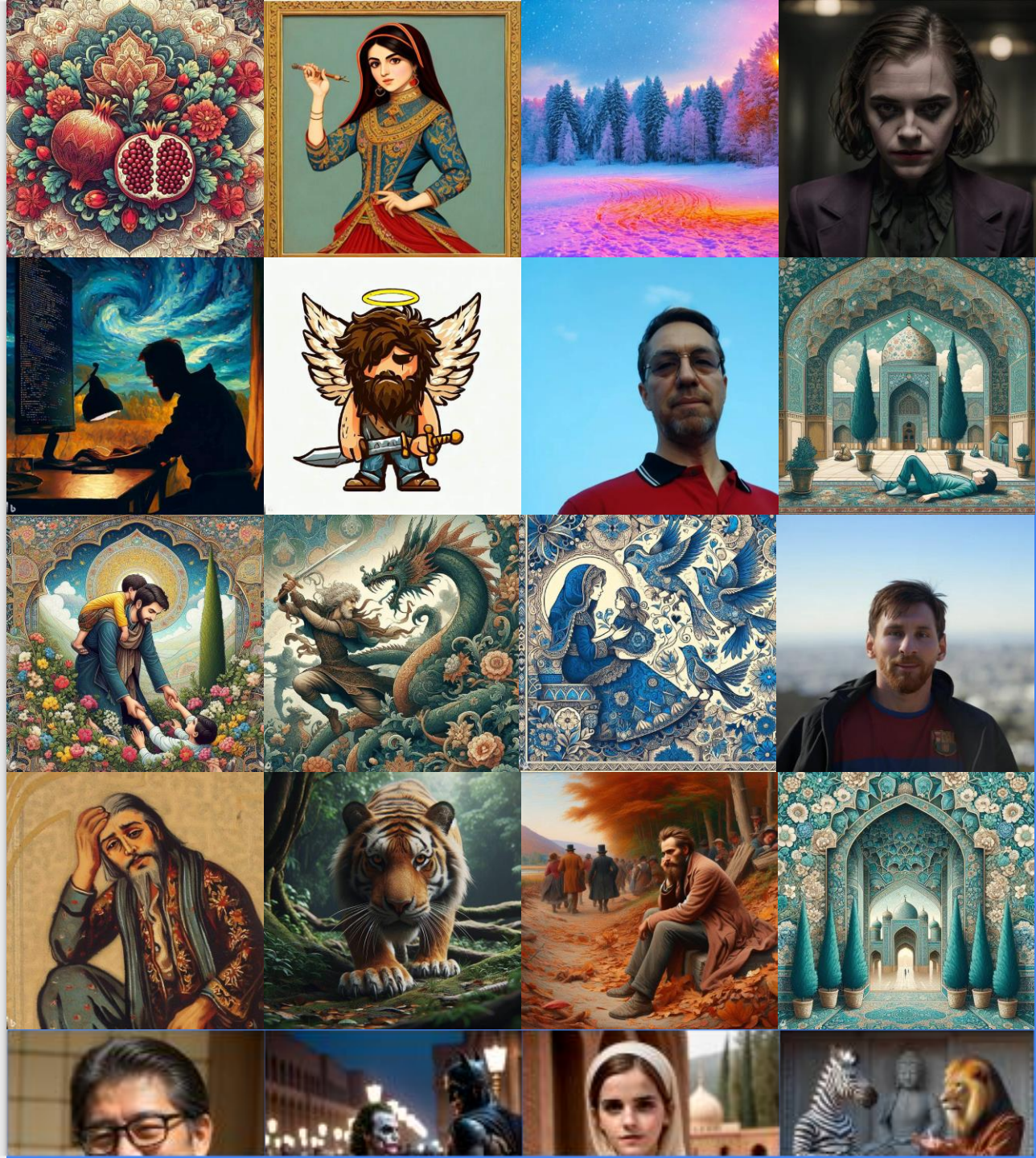
Their own issues: Deep Learning

Different Modals: Image, Text, Speech, Graph, …

What is Gen AI?

creating new contents

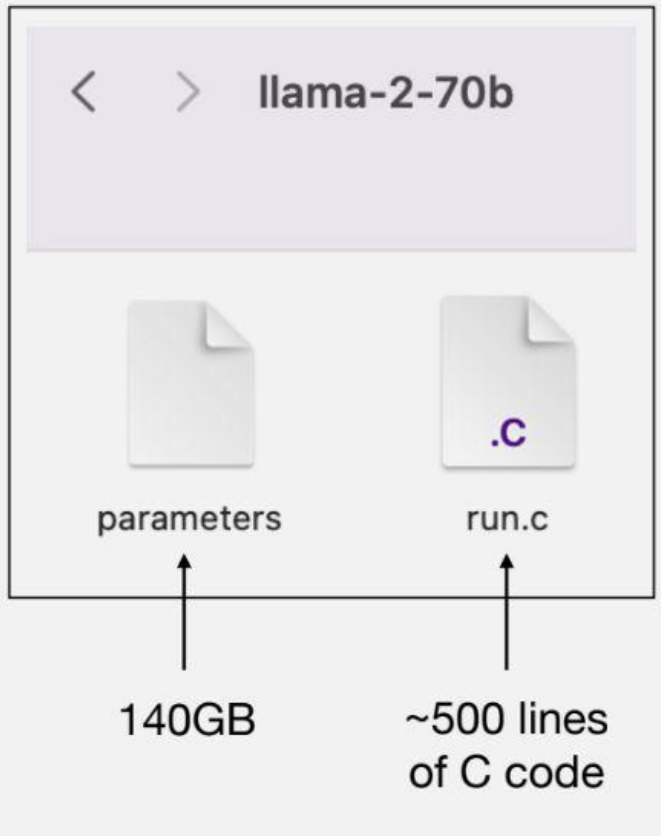


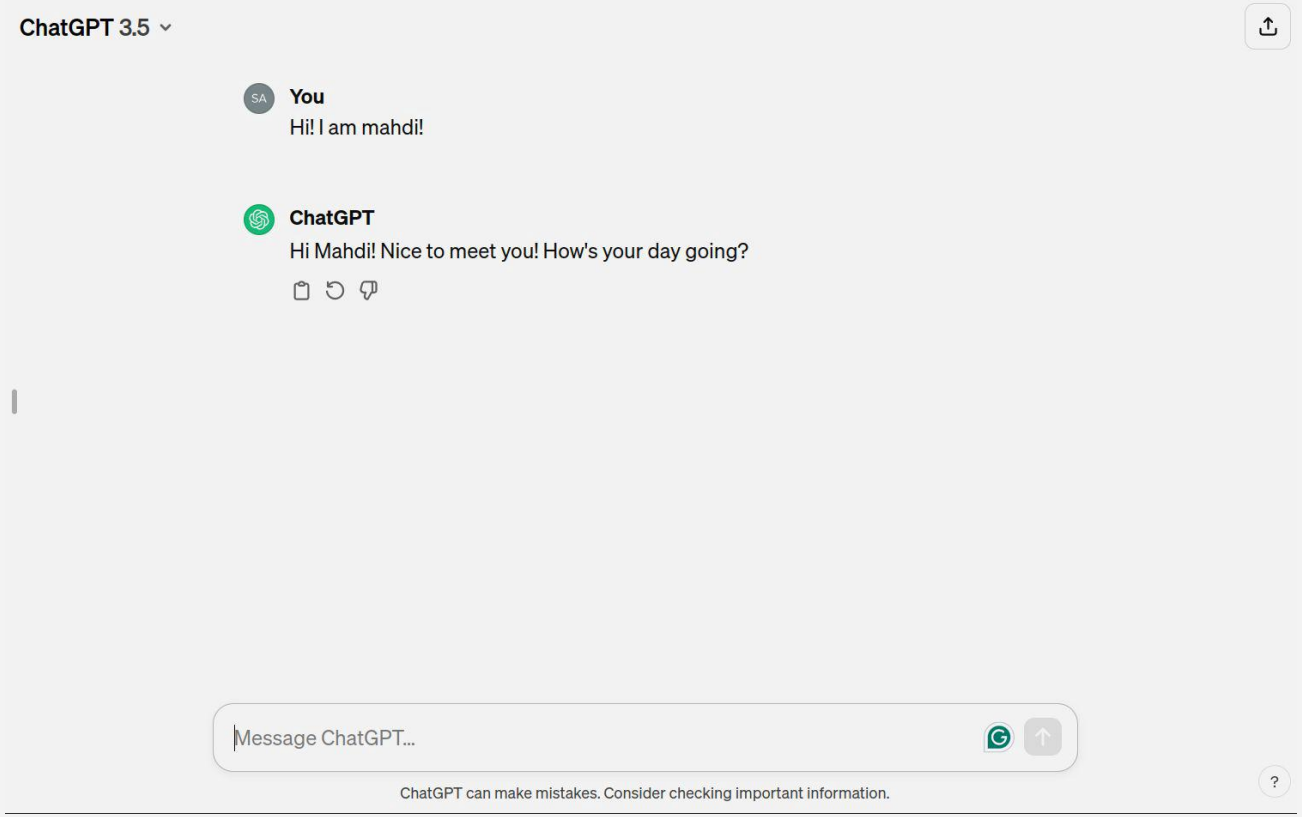


LLM?

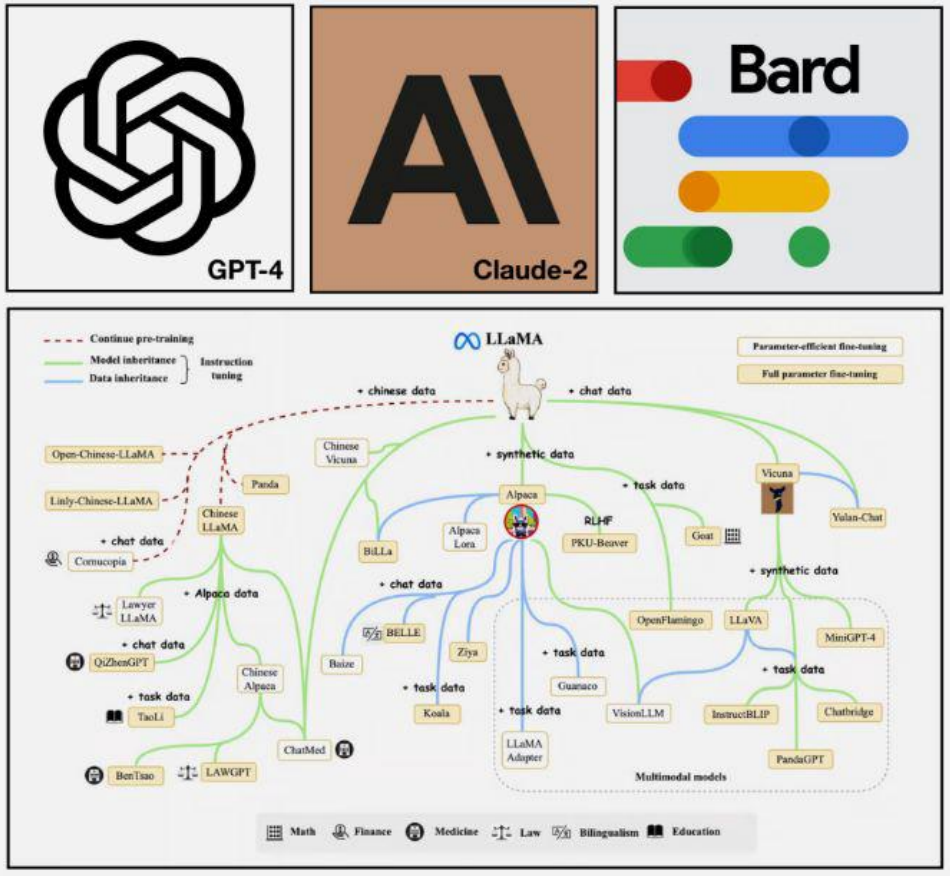
- What is LLM?
- How LLM is Created?
- What LLM Does?

What is LLM?


llama-2-70b
.c
parameters
run.c
140GB
~500 lines
of C code


ChatGPT 3.5
SA
You
Hi! I am mahdi!
ChatGPT
Hi Mahdi! Nice to meet you! How's your day going?
Message ChatGPT...
ChatGPT can make mistakes. Consider checking important information.

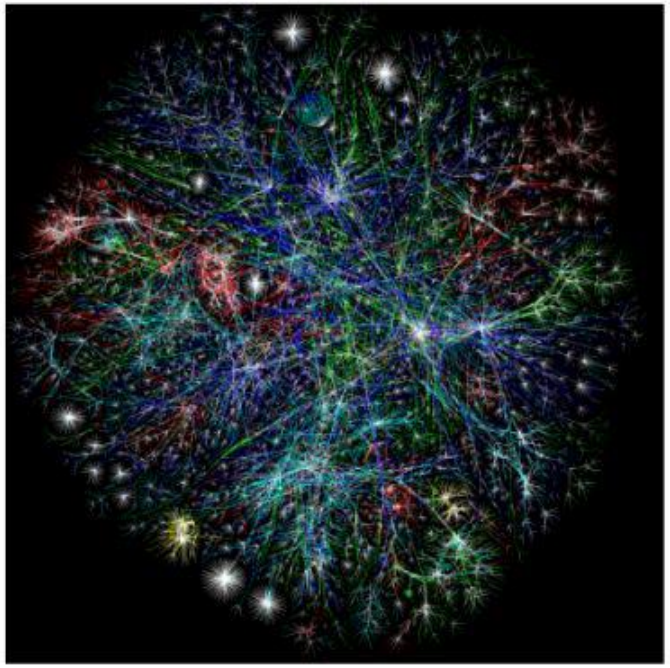
Close-Source LLM vs Open-Source LLM



How is a Large Language Model built?



Phase 1: Train LLM on Lots of text by NTP



Chunk of the internet,
~10TB of text



6,000 GPUs for 12 days, ~$2M
~1e24 FLOPS

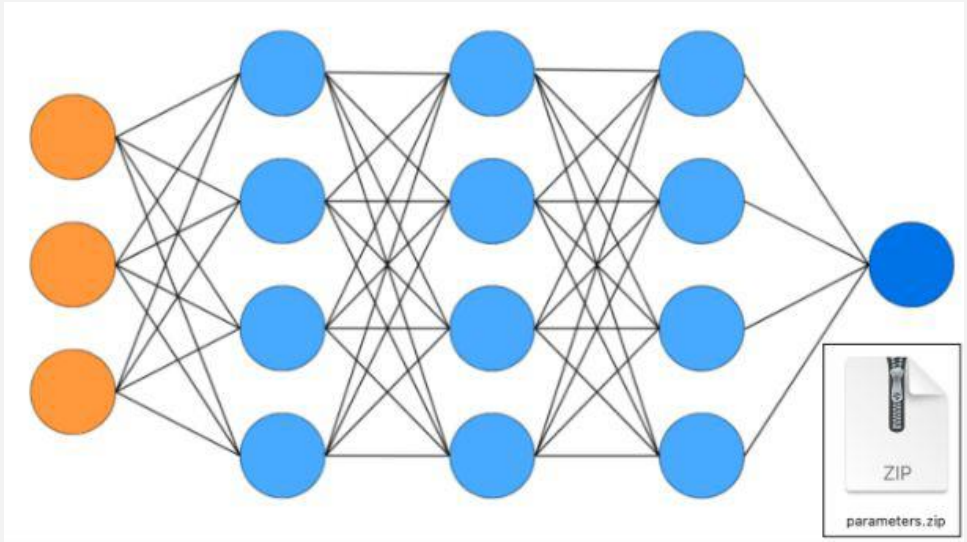


parameters.zip

~140GB file

What you get:

<شروع>



محمدرضا، ۱۳ درصد

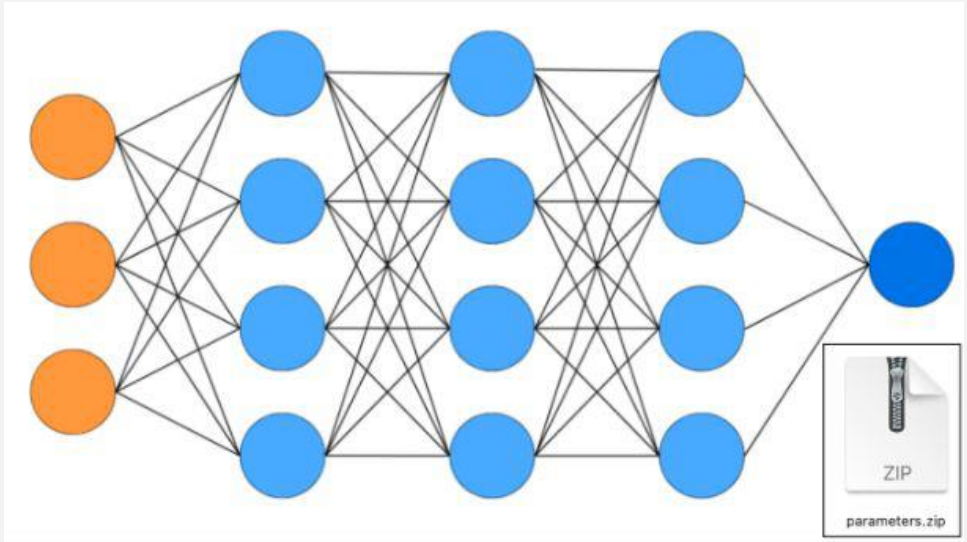
حسین، ۴ درصد

...

بود، ۰ درصد

What you get:

<شروع>
محمدرضا



پسر، ۹۷ درصد

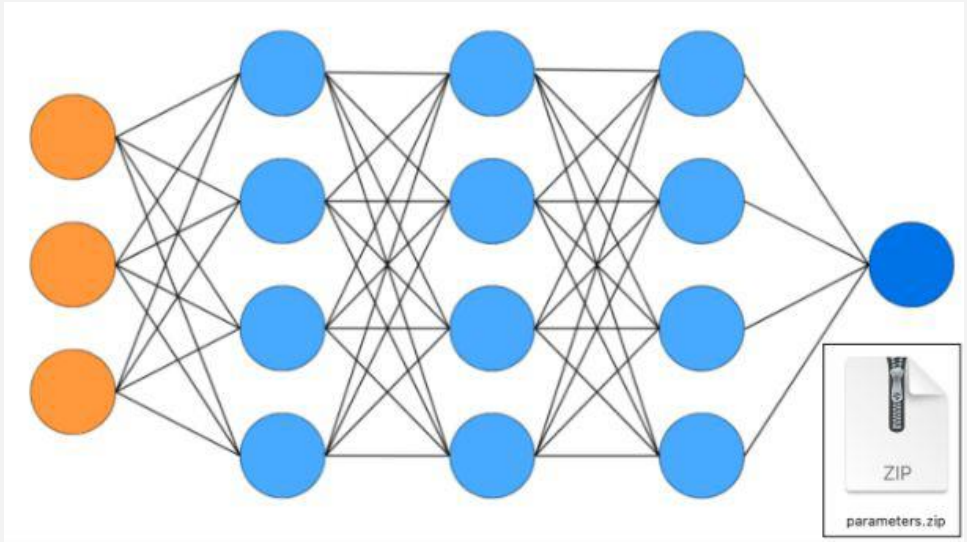
فریدونی، ۲ درصد

...

دختر، ۰ درصد

What you get:

<شروع>
محمدرضا پسر



خوبی، ۹۰ درصد

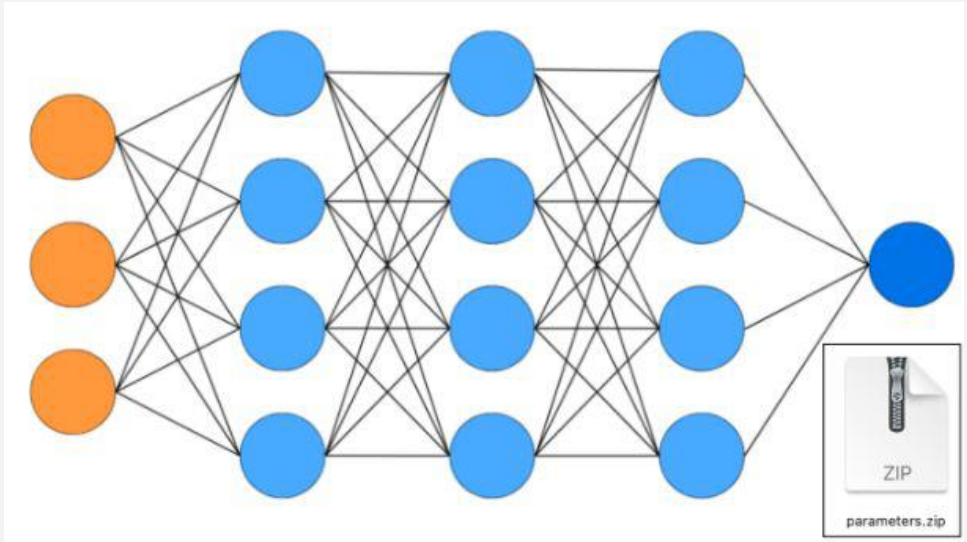
باهوشی، ۵ درصد

...

اوگاندا، ۰ درصد

What you get:

<شروع>
محمدرضا پسر
خوبی



نیست، ۷۲ درصد

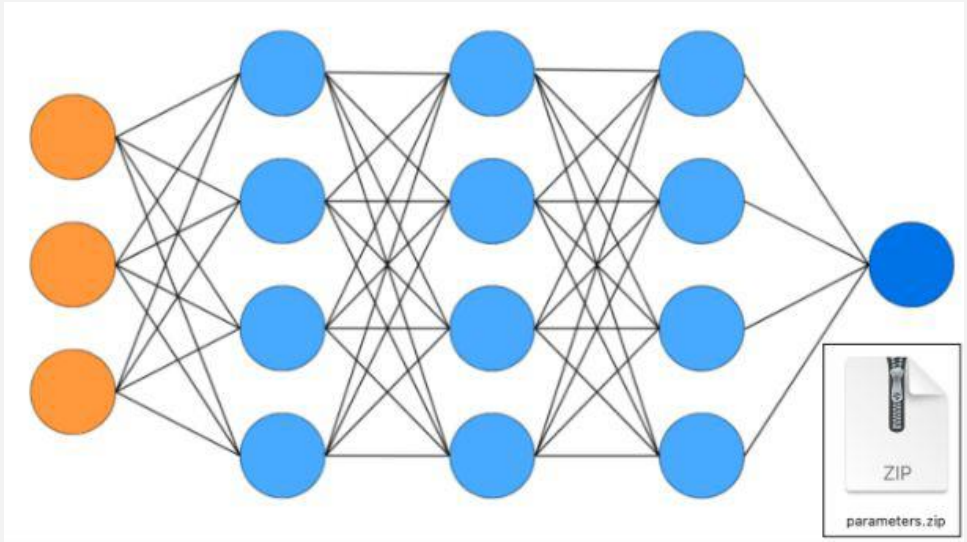
هست، ۱۲ درصد

...

کلنگ، ۰ درصد

What you get:

<شروع>
محمدرضا پسر
خوبی نیست



<پایان> ، ۴۳
درصد

اما، ۵ درصد

...

زرافه، ۰ درصد

So By Training NSP on Vast Amount of Text, Knowledge Emerges!

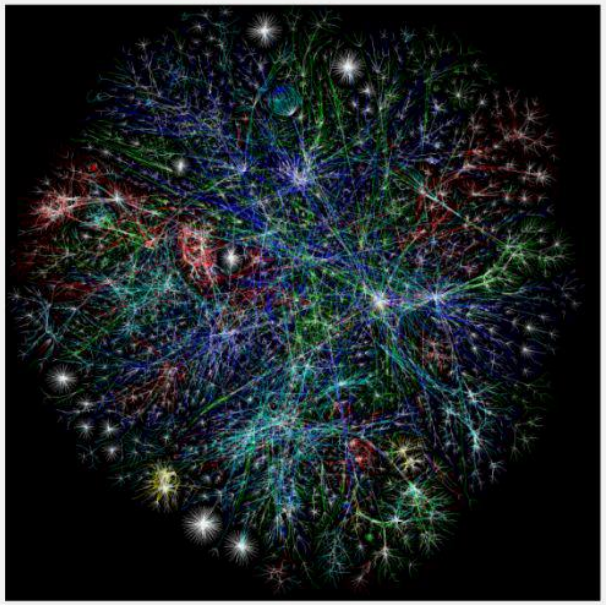
So By Training NSP on Vasts Amount of Text, Knowledge Emerges!

But it’s not Enough!

We Want Assistant: Collecting Instruction Based Dataset

Training the Assistant

(Just swap the dataset, then continue training)



written by people:

<USER>

Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>

"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions…

~100K conversations

We Want Assistant: Finetune Parameters on Instruction Dataset

After finetuning you have an Assistant:

<USER>

Can you help me with this code? It seems like there is a bug.
print("hello world)

<ASSISTANT>

It looks like you forgot to close the string passed to the function print. You have to add a closing quote to properly terminate the string. Here is the corrected function:

print("hello world")

Let me know if I can help with anything else!

What is missing?

🔻 Misalignment: When the training objective does not capture the desiderata we want from models

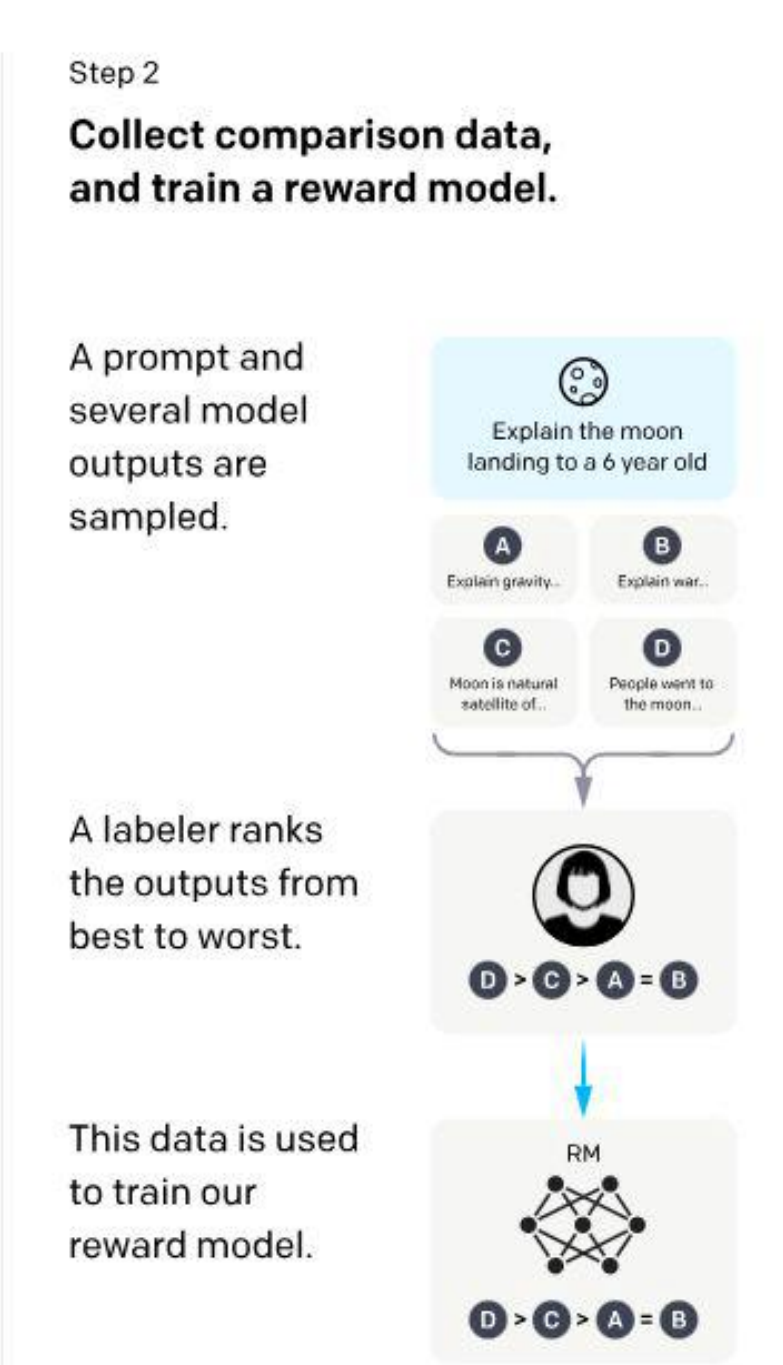
❓ Helpful: should help the user solve their task according to the instructions.

❓ Honest: should express uncertainty when the model doesn't know the answer, instead of hallucinating a wrong answer.

❓ Harmless: should not cause physical, psychological, or social harm to people or the environment.

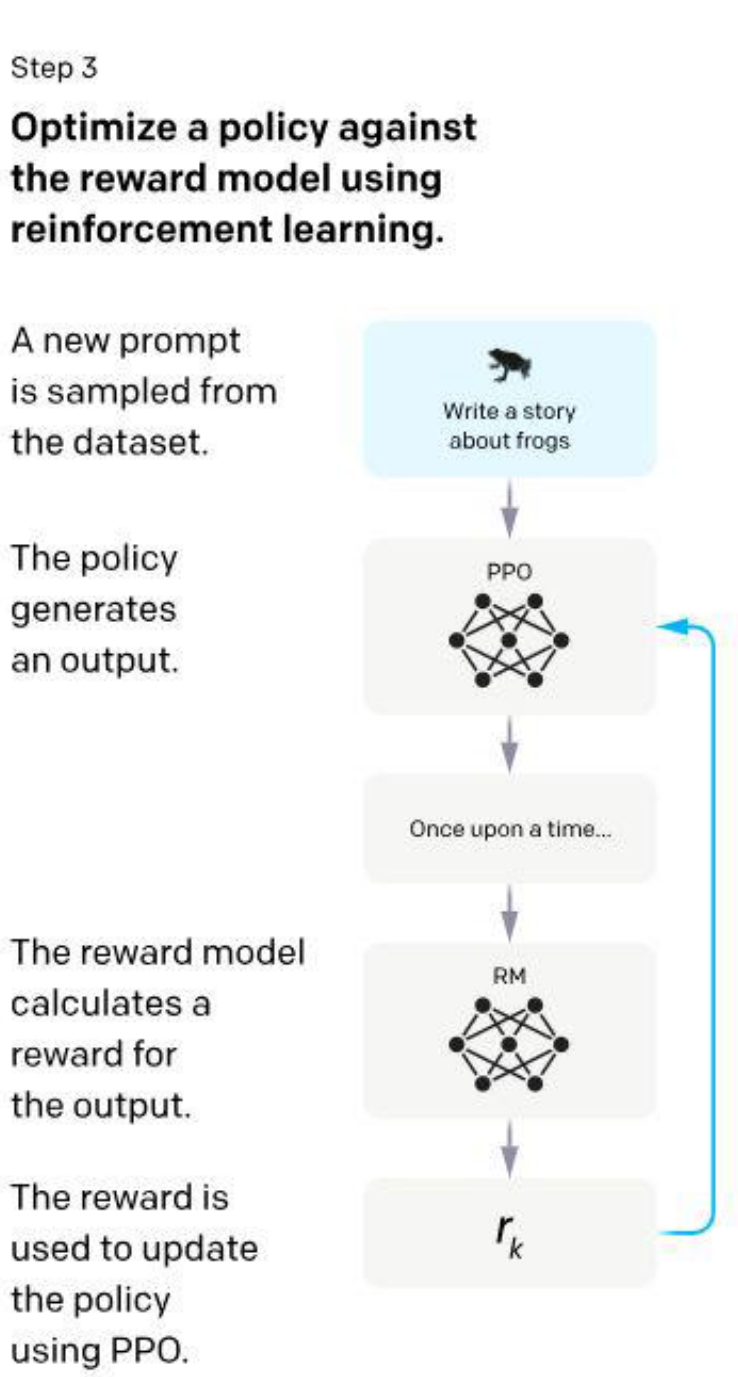
RLHF: Change the Loss

- Use reward signal and Reinforcement Learning to fine tune the model.
- Need scored data: (instruction, answer, score)



RLHF: Change the Loss

- Use reward signal and Reinforcement Learning to fine tune the model.
- Need scored data: (instruction, answer, score)
- Fine-Tune LLM using Proximal Policy Optimization on Reward Function



What LLM Does?

How to behave with your llm?

Where LLM goes?

LLM Leaderboards

Arena Elo | Full Leaderboard

Three benchmarks are displayed: **Arena Elo**, **MT-Bench** and **MMLU**.

- Chatbot Arena - a crowdsourced, randomized battle platform based on human preference votes.
- MT-Bench: a set of challenging multi-turn questions. We use GPT-4 to grade the model responses.
- MMLU (5-shot): a test to measure a model's multitask accuracy on 57 tasks.

💻 Code: The MT-bench scores (single-answer grading on a scale of 10) are computed by fastchat.llm_judge. The MMLU scores are mostly computed by InstructEval. Higher values are better for all benchmarks. Empty cells mean not available.

🤖 Model	⭐ Arena Elo	📈 MT-bench	📚 MMLU	Organization	License
GPT-4-1106-preview	1254	9.32		OpenAI	Proprietary
GPT-4-0125-preview	1253			OpenAI	Proprietary
Bard (Gemini Pro)	1218			Google	Proprietary
GPT-4-0314	1191	8.96	86.4	OpenAI	Proprietary
GPT-4-0613	1164	9.18		OpenAI	Proprietary
Mistral Medium	1152	8.61	75.3	Mistral	Proprietary
Claude-1	1150	7.9	77	Anthropic	Proprietary
Qwen1.5-72B-Chat	1147	8.61	77.5	Alibaba	Qianwen LICENSE
Claude-2.0	1132	8.06	78.5	Anthropic	Proprietary
Gemini Pro (Dev API)	1122		71.8	Google	Proprietary
Claude-2.1	1120	8.18		Anthropic	Proprietary
Mixtral-8x7b-Instruct-v0.1	1120	8.3	70.6	Mistral	Apache 2.0
GPT-3.5-Turbo-0613	1118	8.39		OpenAI	Proprietary
Gemini Pro	1115		71.8	Google	Proprietary
Yi-34B-Chat	1111		73.5	01 AI	Yi License
Claude-Instant-1	1109	7.85	73.4	Anthropic	Proprietary
GPT-3.5-Turbo-0314	1106	7.94	70	OpenAI	Proprietary
WizardLM-70B-v1.0	1106	7.71	63.7	Microsoft	Llama 2 Community

LLMs are not GOD yet

Recent viral example: “reversal curse”

Q: “Who is Tom Cruise’s mother”?
A: Mary Lee Pfeiffer ✅

Q: “Who is Mary Lee Pfeiffer’s son?”
A: I don’t know ❌ 🤔

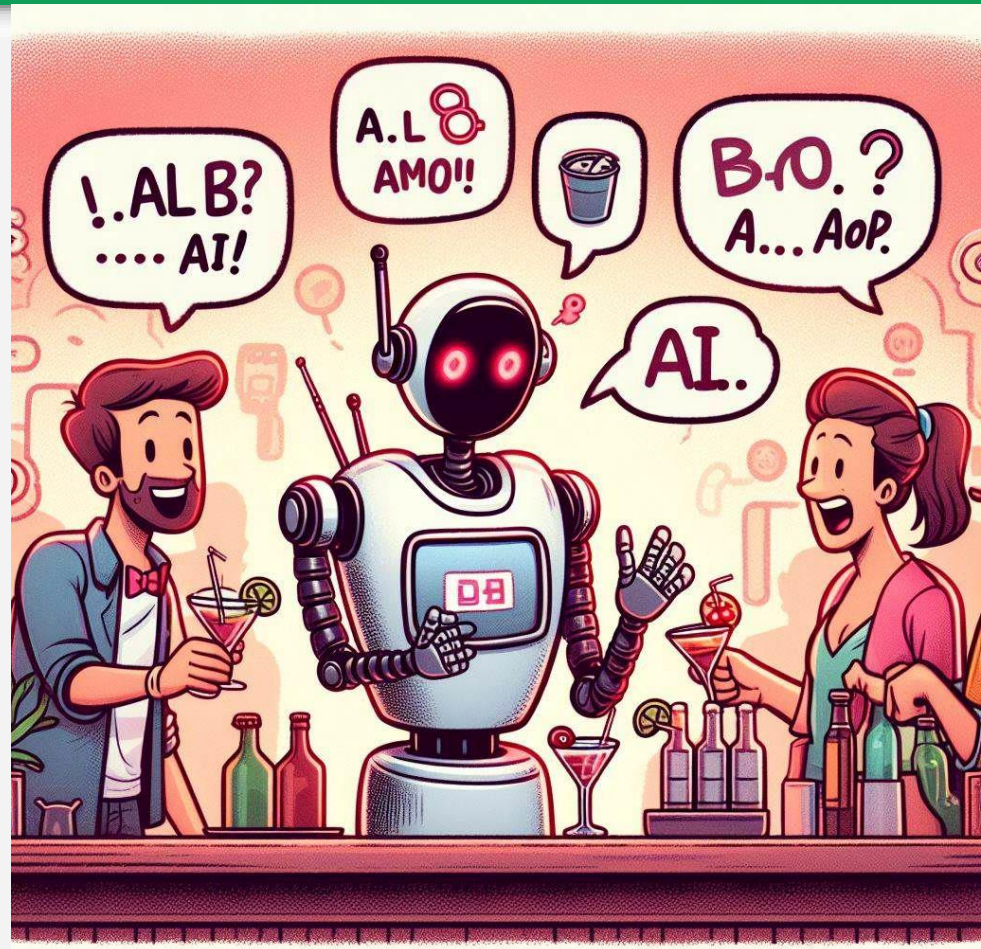
Hallucination!

You
What is the name of wife of mostafa meshkati?

ChatGPT
Mostafa Meshkati, the Iranian-American engineer known for his work in human factors engineering and safety, is married to Dr. Sepideh Modrek.

You must know how to talk to LLM: Prompt Engineering

Prompt Engineering, also known as **In-Context Prompting**, refers to methods for how to communicate with LLM to steer its behavior for desired outcomes without updating the model weights. It is an empirical science and the effect of prompt engineering methods can vary a lot among models, thus requiring heavy experimentation and heuristics.



Prompt Engineering: Few-Shot Demonstration

Zero-Shot Prompting

```
Text: i'll bet the video game is a lot more fun than the film.
Sentiment:
```

few-Shot Prompting

```
Text: (lawrence bounces) all over the stage, dancing, running, sweating, mopping his face and g
Sentiment: positive

Text: despite all evidence to the contrary, this clunker has somehow managed to pose as an actu
Sentiment: negative

Text: for the first time in years, de niro digs deep emotionally, perhaps because he's been sti
Sentiment: positive

Text: i'll bet the video game is a lot more fun than the film.
Sentiment:
```

Prompt Engineering: Instruction Prompting

The purpose of presenting few-shot examples in the prompt is to explain our intent to the model; in other words, describe the task instruction to the model in the form of demonstrations. However, few-shot can be expensive in terms of token usage and restricts the input length due to limited context length. So, why not just give the instruction directly?

```
Definition: Determine the speaker of the dialogue, "agent" or "customer".
Input: I have successfully booked your tickets.
Ouput: agent

Definition: Determine which category the question asks for, "Quantity" or "Location".
Input: What's the oldest building in US?
Ouput: Location

Definition: Classify the sentiment of the given movie review, "positive" or "negative".
Input: i'll bet the video game is a lot more fun than the film.
Output:
```

Prompt Engineering: CoT Prompting

```
Question: Tom and Elizabeth have a competition to climb a hill. Elizabeth takes 30 minut
Answer: It takes Tom 30*4 = <<30*4=120>>120 minutes to climb the hill.
It takes Tom 120/60 = <<120/60=2>>2 hours to climb the hill.
So the answer is 2.
===
Question: Jack is a soccer player. He needs to buy two pairs of socks and a pair of soccer shoe
Answer: The total cost of two pairs of socks is $9.50 x 2 = $<<9.5*2=19>>19.
The total cost of the socks and the shoes is $19 + $92 = $<<19+92=111>>111.
Jack need $111 - $40 = $<<111-40=71>>71 more.
So the answer is 71.
===
Question: Marty has 100 centimeters of ribbon that he must cut into 4 equal parts. Each of the
Answer:
```

copy

LLM are not enough?
The world is changing

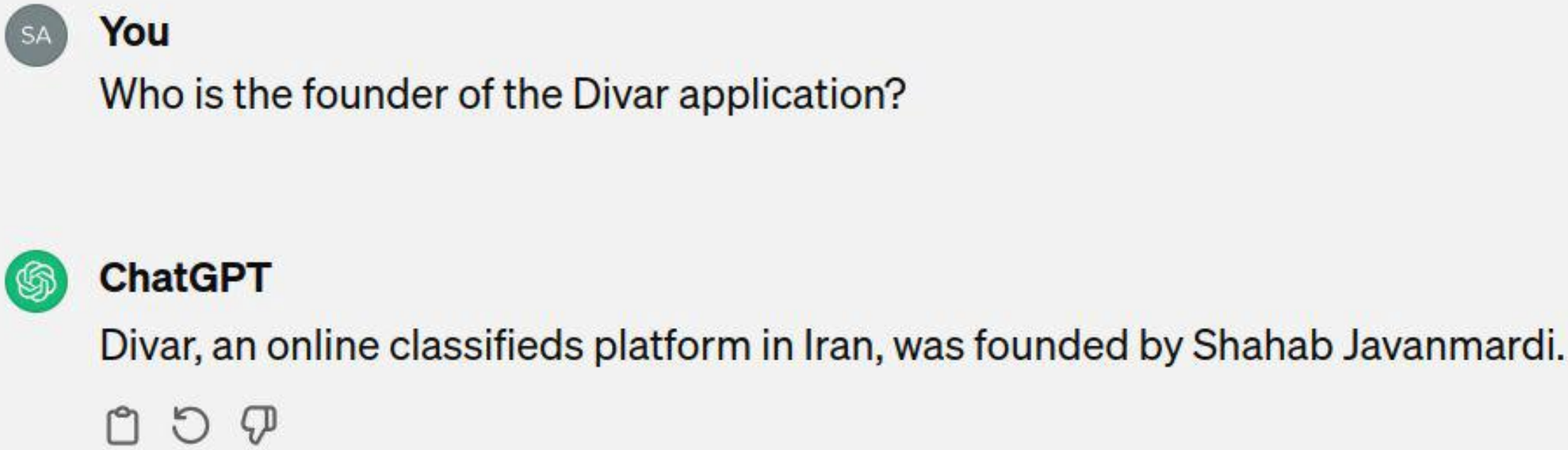
SA **You**

When was the last Israeli-Palestinian war?

ChatGPT

The most recent major conflict between Israel and Palestinian militant groups was the 11-day war in May 2021. This conflict primarily involved Israel and Hamas, the militant group controlling the Gaza Strip. It resulted in significant casualties and destruction before a ceasefire was brokered by Egypt and other international mediators.

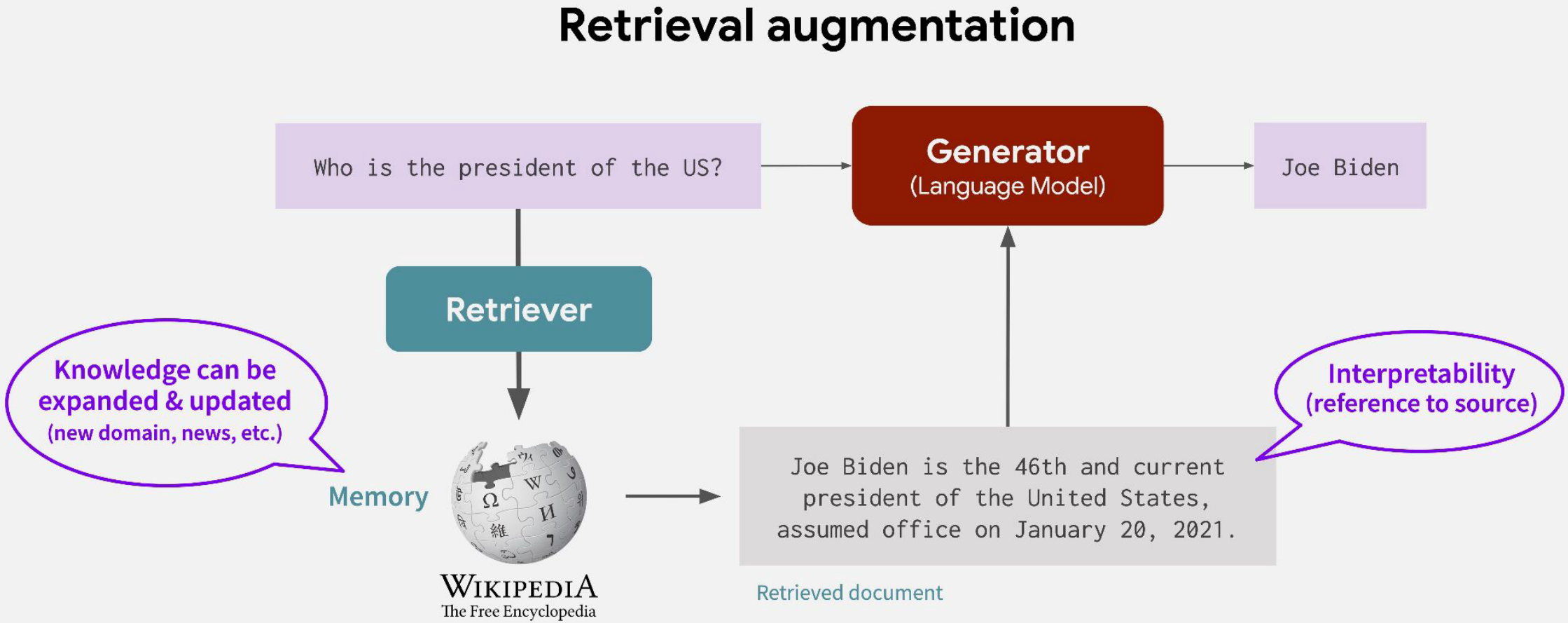
LLM are not enough?
LLM doesn’t know everything



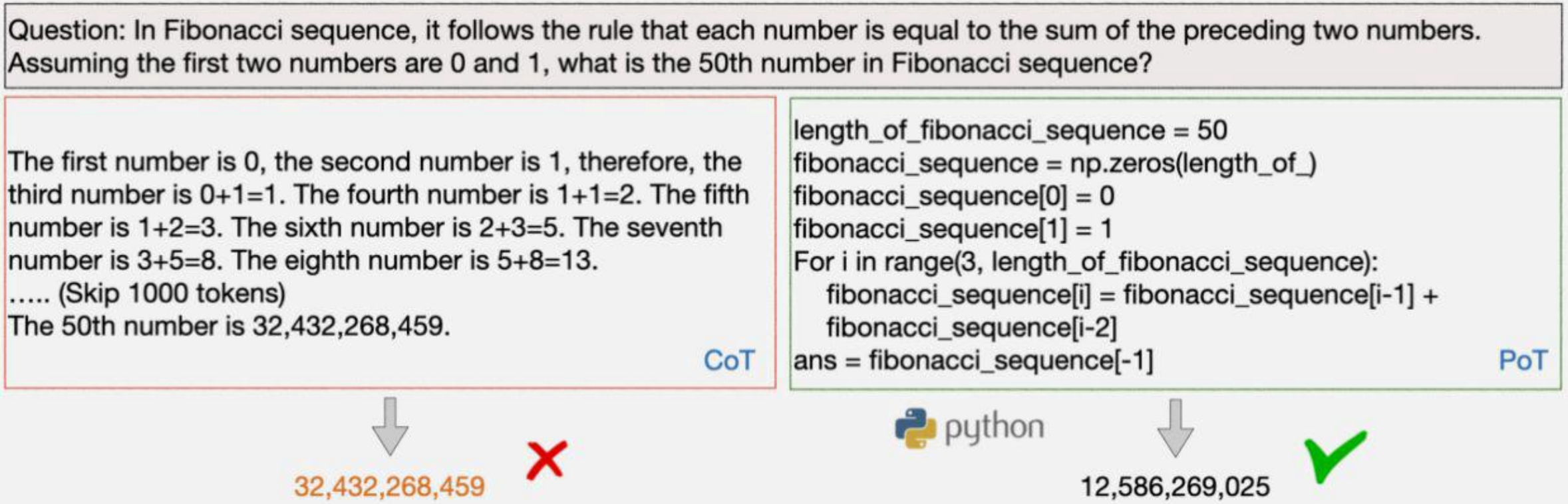
What does human do?



LLMs: Retrieval Augmented Generation



LLMs and External API



LLM is a Modern KNN

It all depends on what data points you choose to input into LLM.

	SST-2	AGNews	R8	R52	MR	Average
Supervised Methods						
RoBERTa-Large (Liu et al., 2019)	95.99	95.55	97.76	96.42	91.16	95.38
DeBERTa (He et al., 2020)	94.75	95.32	98.33	96.32	90.19	94.99
RoBERTa-GCN (Lin et al., 2021)	95.80	**95.68***	98.2	96.1	89.7	95.10
XLNet (Yang et al., 2019)	**96.10***	95.55	-	-	-	-
VLAWE (Ionescu and Butnaru, 2019)	-	-	-	-	**93.3***	-
GCN-SB (Zeng et al., 2022)	-	-	**98.53***	**96.35***	87.59	-
Zero-shot Setting						
Vanilla (Brown et al., 2020)	91.55	90.72	90.19	89.06	88.69	90.04
CoT (Kojima et al., 2022)	92.11	91.25	90.48	91.24	89.37	90.89
CARP	93.01	92.60	91.75	91.80	89.94	91.82
Few-shot Setting (k=16)						
Random Sampler						
Vanilla (Brown et al., 2020)	92.36	91.74	91.58	91.56	89.15	91.28
CoT (Kojima et al., 2022)	94.56	95.02	92.49	92.03	89.91	92.80
CARP	96.20	95.18	97.60	96.19	90.03	95.04
SimCSE kNN-Sampler						
Vanilla (Brown et al., 2020)	93.90	93.50	94.36	92.40	89.59	94.05
CoT (Kojima et al., 2022)	94.21	94.28	95.07	92.98	90.27	93.69
CARP	95.69	95.25	97.83	96.27	90.74	95.16
FT kNN-Sampler						
Vanilla (Brown et al., 2020)	94.01	94.14	95.57	95.79	90.90	94.08
CoT (Kojima et al., 2022)	95.48	94.89	95.59	95.89	90.17	94.40
CARP	96.80	95.99	98.29	96.82	91.90	95.97
CARP (WP Vote)	97.39	96.40	98.78	96.95	92.39	96.38

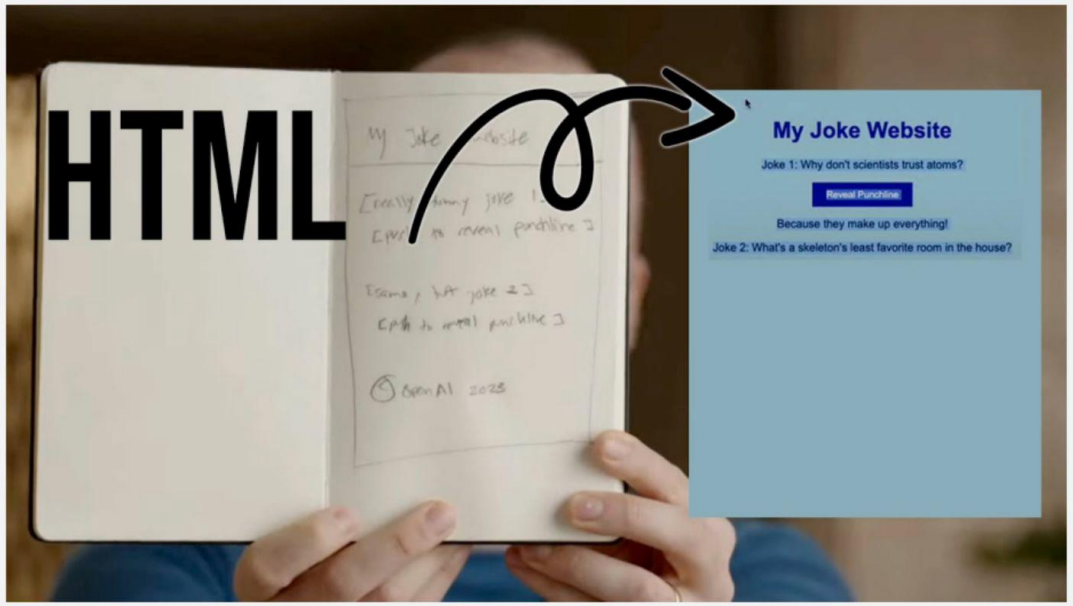
Where LLM goes?



Multi-Modal LLMs

Vision

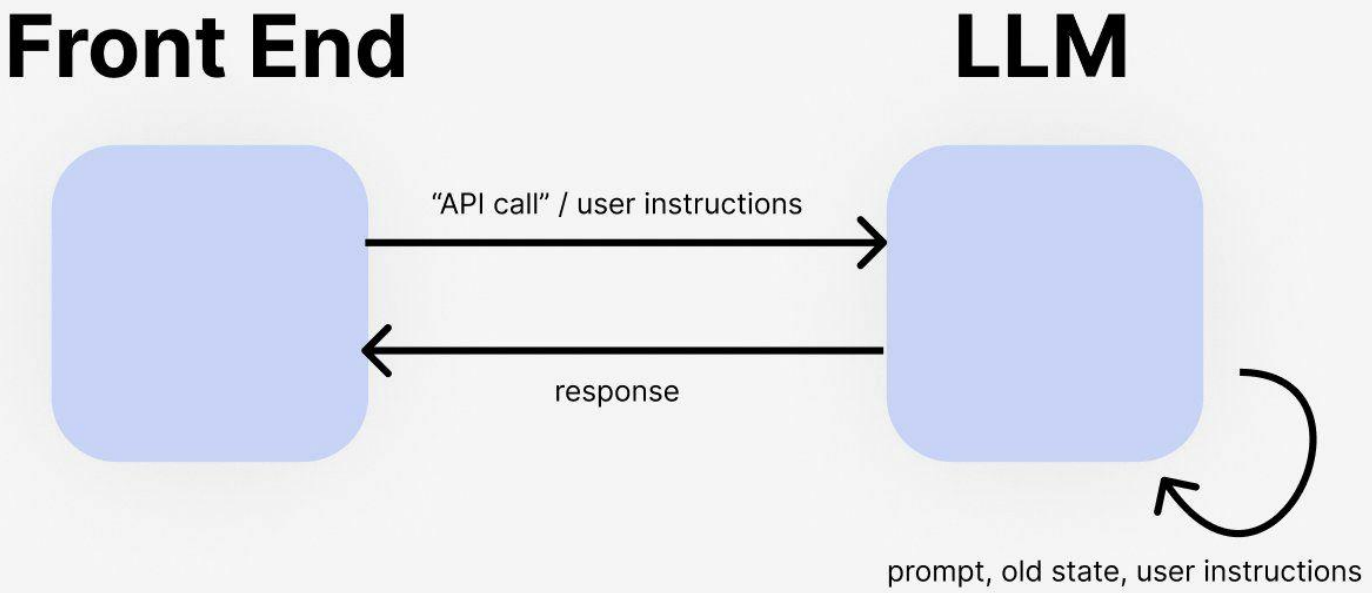
Can both see, and generate images



Example: Take a sketch of an idea and generate a working website.

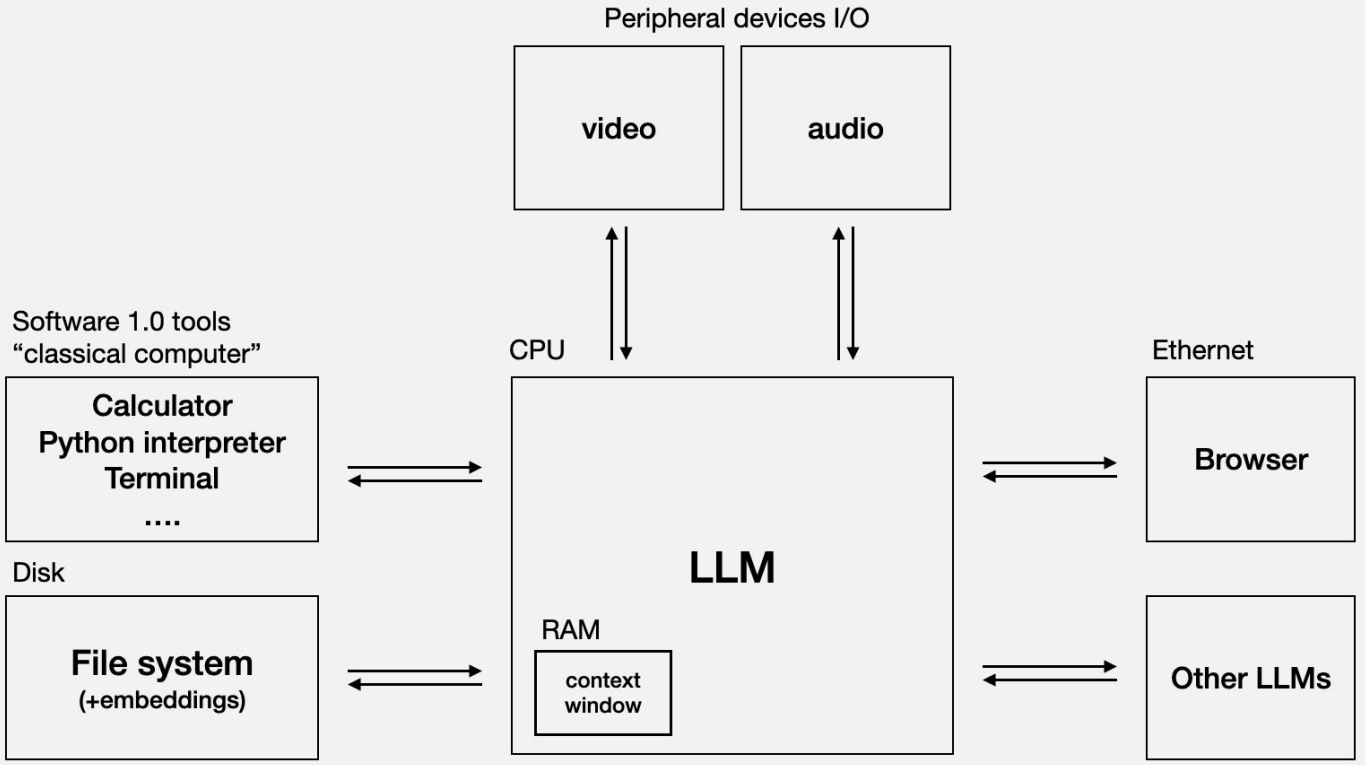
LLM as Backend





https://github.com/RootbeerComputer/backend-GPT

LLM as OS



- It can read and generate text
- It has more knowledge than any single human about all subjects
- It can browse the internet
- It can use the existing software infrastructure (calculator, Python, mouse/keyboard)
- It can see and generate images and video
- It can hear and speak, and generate music
- It can think for a long time using a System 2
- It can “self-improve” in domains that offer a reward function
- It can be customized and finetuned for specific tasks, many versions exist in app stores
- It can communicate with other LLMs



Andrej Karpathy

@karpathy

The hottest new programming language is English